UNITED INTERNATIONAL UNIVERSITY

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Final Project Report

Accessibility & Ethical AI for Visually Impaired

Human Computer Interaction (HCI)

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Abstract

Artificial intelligence (AI) can support information access, communication, and environmental awareness, but its usefulness depends on accessible and dependable interaction. This project investigates barriers experienced by a user with visual impairment and translates the findings into an accessibility-focused interactive prototype. A qualitative formative interview explored assistive technology use, speech interaction, explanation preferences, and independent navigation. Four themes emerged: poor integration with established assistive workflows, difficulty recovering from speech-recognition errors, excessive listening effort for lengthy explanations, and the need for contextual support in unfamiliar environments.

These findings informed three proposed features: navigation and contextual assistance, a voice-based AI query system, and adaptive audio explanations. Paper sketches and peer feedback supported iterative development of a Figma prototype. An assisted evaluation with one visually impaired participant identified difficult control discovery, inconsistent interaction patterns, and limited feature access. The subsequent redesign introduced broader touch interaction, contextual gesture controls, spoken confirmation, explanation replay and saving, and guided onboarding.

The study contributes a traceable account of how qualitative accessibility concerns can inform interaction design. Its evidence supports design requirements and identifies usability problems; it does not establish improvements in recognition accuracy, cognitive load, or safe independent mobility. The prototype simulates interaction rather than demonstrating a deployed AI or navigation system. Further work should evaluate a functioning implementation with a more diverse participant group, screen-reader compatibility, explicit privacy controls, and controlled mobility testing.

Keywords: accessibility; visual impairment; human–computer interaction; ethical AI; voice interaction; audio explainability; user-centred design.

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Abbreviations

AI	Artificial intelligence
AT	Assistive technology
HCI	Human–computer interaction
JAWS	Job Access With Speech
NVDA	NonVisual Desktop Access
OCR	Optical character recognition
TTS	Text-to-speech
WCAG	Web Content Accessibility Guidelines
XAI	Explainable artificial intelligence

Chapter 1

Introduction and Problem Statement

1.1 Background

AI interfaces support activities such as asking questions, interpreting content, and obtaining contextual information. For blind and low-vision users, these functions can be valuable when they work with screen readers, speech input, keyboard navigation, and established assistive practices. Access depends on the complete interaction: discovering a feature, providing input, understanding feedback, correcting mistakes, and deciding whether to trust a response.

A system can provide a technically detailed answer while still imposing substantial effort on a person who receives it through audio. Similarly, a speech interface may appear accessible but become frustrating when recognition errors can only be corrected through visual controls. Accessibility therefore requires attention to the interaction process as well as the information produced.

This project treats accessibility as part of ethical AI design. People should be able to understand system states, control their data, recover from errors, and decide when assistance is appropriate. The design focuses on voice interaction and adaptable audio while recognizing that preferences differ and that speech cannot serve every user or context.

1.2 Problem Statement

The motivating problem is a mismatch between the interaction assumptions of AI interfaces and the participant's non-visual workflows. The formative account describes four connected difficulties:

1. AI features do not always fit established assistive technology routines.
2. Misrecognized speech creates repeated correction effort when visual checking is unavailable.
3. Long spoken responses are difficult to review and retain without useful playback controls.
4. Unfamiliar physical environments create a need for timely contextual information.

These concerns motivate a design that reduces dependence on visually locating controls, makes recognition repair explicit, and offers user-selected explanation detail. Navigation assistance is explored as a concept requiring substantially more technical and safety validation than an interface prototype can provide.

1.3 Research Objectives

The project aims to identify accessibility barriers in an individual's experience of AI-assisted interaction and use those findings to develop and evaluate an accessible interface concept. Its objectives are to:

- document needs relating to assistive technology, speech, explanations, and navigation;
- translate qualitative findings into practical design requirements;
- develop an interactive prototype with voice-oriented workflows and adaptive audio;
- identify usability problems through feedback and observation; and
- describe ethical requirements and the limits of the available evidence.

1.4 Research Questions

RQ1.
What barriers arise when a visually impaired user incorporates AI interfaces into existing assistive technology workflows?

RQ2.
How do speech recognition and spoken explanation design affect the reported interaction experience?

RQ3.
How can participant feedback inform more accessible controls, confirmation, and explanation delivery?

RQ4.
What requirements and unresolved risks arise when proposing AI-based contextual navigation assistance?

1.5 Scope and Contribution

This is an exploratory HCI design case study covering a formative interview, iterative interface design, an assisted prototype evaluation, and a resulting redesign. Its contributions are a synthesis of reported barriers, a mapping from those barriers to features, and a critical account of usability limitations.

The deliverable is a Figma interaction prototype. Environmental sensing, smart-glasses integration, speech recognition, and AI response generation are proposed capabilities, not validated implementations. No benchmark, controlled comparison, or field trial is documented. Conclusions are therefore framed as design insights rather than population-level or causal findings.

Chapter 2

Related Work and Design Foundations

2.1 User-Centred Explainability

Liao, Gruen, and Miller investigate the gap between XAI algorithms and practical user needs through interviews with design practitioners. Their question-based approach treats explanation design as a user-experience problem [1]. For this project, an explanation should help the user answer a task-relevant question. A longer response alone does not establish understanding. The proposed interface therefore separates an initial summary from additional detail available on request.

2.2 Description, Context, and Representation

Bennett and colleagues examine image-description practices and the consequences of representing race, gender, and disability. Their work highlights contextual preferences and the possibility of harmful representation in human- and AI-generated descriptions [2]. Applied here, descriptions should avoid unsupported sensitive inferences, distinguish observation from interpretation, and allow users to control the detail they receive. This is particularly relevant to a future camera-based assistance feature.

2.3 Accessibility Standards

WCAG 2.2 provides testable criteria for accessible web content, including keyboard access, meaningful labels, status messages, and alternatives to certain gestures [3]. These offer a useful interface baseline. Applying them to this concept suggests accessible control names, predictable focus order, perceivable state changes, and alternatives to custom gestures. The prototype has not undergone a conformance audit; its appearance and voice-oriented concept do not establish compliance. A native mobile implementation would also need platform-specific accessibility testing.

2.4 Responsible AI and Risk Management

The NIST AI Risk Management Framework organizes risk work around governance, context mapping, measurement, and management. It considers reliability, safety, transparency, privacy, and fairness among the characteristics of trustworthy AI [4]. Its practical value here is to connect design decisions with specific risks and verification activities. A navigation prompt, for example, needs evidence about failure conditions as well as understandable wording.

2.5 Project Position

The project connects these foundations through three design problems: how the user initiates an action, how the system communicates what it understood, and how the user can inspect or challenge its response. The formative interview grounds these problems in a particular experience; the prototype explores possible responses.

This report does not claim that accessibility is absent from all AI policies or that current systems are uniformly inaccessible. Its narrower contribution is to show why broad principles must be translated into interaction requirements and evaluated with intended users.

Chapter 3

Research Methodology

3.1 Study Design

A qualitative, user-centred approach was used in two stages. The formative stage explored everyday practices and unmet needs through a semi-structured interview. The design and evaluation stage translated these accounts into paper sketches and a Figma prototype, obtained peer feedback, and collected feedback during an assisted walkthrough.

The purpose was exploratory: to identify problems and refine a design concept. The study was not designed to estimate prevalence, compare algorithms, or establish effectiveness. Findings are based on reported experience and qualitative observation.

3.2 Participant and Recruitment

The formative study used purposive sampling to recruit one visually impaired participant experienced with assistive technologies. The source account identifies JAWS, NVDA, VoiceOver, Dragon Dictation, and OpenBook OCR among the tools discussed. Her direct exposure to generative AI was limited, although she was familiar with AI applications through professional and community contacts.

The detailed evaluation account documents one visually impaired female participant working in education. She regularly used screen readers, smartphone accessibility features, and audio-based tools, with relatively limited chatbot experience. The records do not establish whether the formative and evaluation participants were the same person; this report does not infer a total number of unique participants across stages.

Reporting clarification. The original abstract refers to three evaluation users, whereas the detailed participant subsection describes one. This report follows the more specific documented account and treats the evaluation as a single-participant study. No additional participants or sessions are inferred.

3.3 Formative Data Collection

An in-person, semi-structured interview was conducted at the participant's workplace and lasted approximately 60–90 minutes. Four topic areas guided the discussion:

Contextual inquiry.
Everyday assistive technologies, routines, and AI experience.

Usability.
Input problems, feature access, and interface navigation.

Explainability.
Spoken responses, information volume, and preferred explanation styles.

Design exploration.
Expectations and suggestions for future accessible assistance.

The flexible format allowed clarification and follow-up questions. The participant's descriptions provided the main basis for interpreting her needs. Assumptions about eye contact or other sight-dependent behaviour are not used as evidence of comprehension.

3.4 Qualitative Analysis

The source report describes four analytic stages: becoming familiar with interview material, identifying initial codes, grouping related codes, and reviewing themes to inform design. The analysis resulted in themes concerning assistive technology integration, speech recognition, explanation length, and navigation.

The report does not provide a full transcript, codebook, coding examples, independent coding procedure, or an audit trail linking each interpretation to an excerpt. The themes should therefore be understood as the project team's qualitative synthesis. This revision paraphrases the available findings and introduces no invented participant quotations.

3.5 Prototype Development and Evaluation

Paper sketches explored the three feature areas. Peer review informed revisions to clutter, confirmation procedures, and interaction flow. A higher-fidelity Figma prototype was evaluated in a session lasting approximately 30–45 minutes. The team assisted the participant in understanding the prototype, observed interaction, and invited feedback.

Three tasks addressed simulated navigation, voice queries, and audio explanations. The assisted walkthrough could reveal problems with discovery and conceptual interaction. It could not establish real recognition accuracy, environmental detection performance, independent task completion, or safe mobility.

3.6 Ethical Reporting and Data Handling

Participant identity is omitted. The available material does not document formal consent, recording permission, institutional ethics review, compensation, or a retention policy. None of these is claimed to have occurred.

Future sessions should use accessible study information and consent, separate recording permission, and a clear withdrawal process. Recordings and transcripts should be minimized, access controlled, and retained only for an agreed period. Assistance during tasks should be recorded because it affects the interpretation of usability and independence.

Chapter 4

Formative Findings and Requirements

4.1 Fit with Assistive Technology Workflows

The participant described established methods for reading, writing, communication, and information access using assistive technologies. AI was less integrated into those routines. Familiarity with AT should therefore not be confused with familiarity with a new AI interface.

The design implication is interoperability. A future implementation should work with existing screen-reader navigation and expose meaningful names, states, and focus behaviour. Voice control should complement these workflows rather than require the user to abandon them.

4.2 Speech Recognition and Error Recovery

The participant reported cases in which similar-sounding words were misunderstood. Repeated correction attempts created additional effort when the recognized text could not be visually checked. The issue is both recognition quality and accessible repair.

The proposed response is a spoken read-back with opportunities to confirm, correct, or cancel. Confirmation introduces listening time, so a future implementation should investigate when it is necessary and whether users want control over it. The project provides no measured recognition error rate.

4.3 Listening Effort and Explanation Control

The participant preferred concise explanations divided into manageable units. Audio unfolds over time; without useful controls, revisiting a point may require replaying more content than necessary. This is an interface concern rather than evidence that visual impairment implies reduced cognitive ability.

The design offers a short summary, optional detail, replay, and saving to support review and control. Effects on comprehension or cognitive workload remain questions for subsequent measurement.

4.4 Contextual Support and Autonomy

Familiar environments were described as easier to navigate than unfamiliar places. The participant expressed interest in obstacles, steps, directions, objects, and places to sit. The underlying need was greater control over movement and timely contextual information.

This motivated the navigation concept, but a stated need does not establish feasibility. Camera-

based assistance would need to handle occlusion, poor lighting, changing obstacles, uncertainty, and failures. Its role should remain supplementary until engineering and supervised evaluation establish appropriate conditions of use.

4.5 Traceability from Findings to Design

Table 4.1 connects findings to prototype responses and remaining validation needs.

Table 4.1: Findings, responses, and unresolved verification needs.

Reported barrier	Design response	Verification needed
Poor fit with AT routines	Voice access alongside accessible controls	Screen-reader and keyboard task testing
Speech misrecognition	Read-back, confirmation, correction, cancellation	Recognition and repair performance with real speech
Lengthy explanations	Summary/detail modes, replay, saved history	Comprehension and workload evaluation
Unfamiliar surroundings	Proposed contextual audio guidance	Sensor, latency, hazard, and supervised mobility testing

Chapter 5

System Design and Prototype

5.1 Design Principles

The concept follows five principles: accessible initiation, low listening burden, explicit user control, predictable interaction, and adaptable detail. Voice is a primary option, but implementation should preserve accessible non-speech alternatives for noisy environments, privacy-sensitive situations, and people who cannot or prefer not to speak.

Listening, processing, awaiting confirmation, responding, and failing are distinct states. Each requires appropriate feedback so silence does not force the user to guess what happened.

5.2 Conceptual Architecture

The proposed architecture comprises input, processing, and output layers. Figure 5.1 summarizes the functional flow. A future contextual assistance implementation would additionally require environmental sensors, localization where appropriate, and a policy governing when uncertain guidance must be withheld. These dependencies are outside the demonstrated Figma interaction.

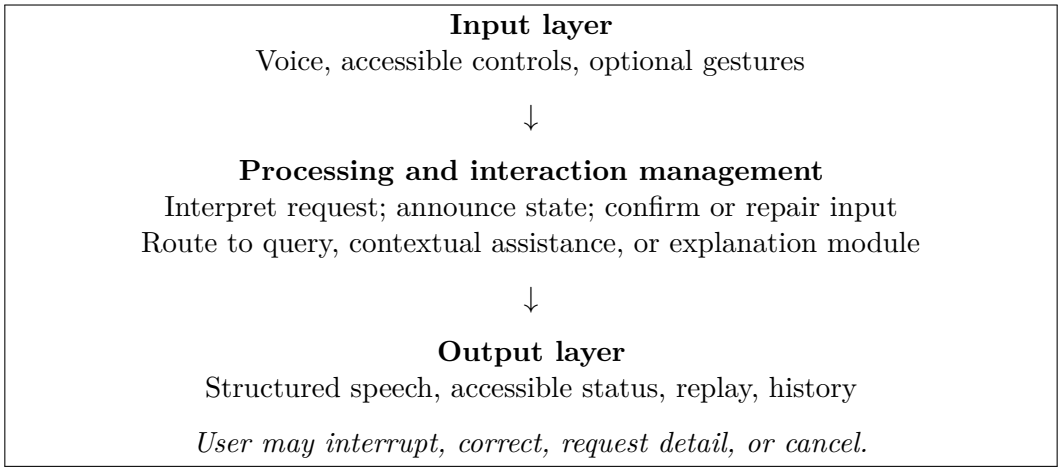


Figure 5.1: Proposed functional architecture, not an implemented AI pipeline.

5.3 Navigation and Contextual Assistance

The navigation concept responds to the participant's interest in environmental information. The original design proposes camera-equipped smart glasses for obstacle descriptions, directional guidance, step warnings, and object finding. These are intended capabilities; the report contains no evidence of working sensor integration, a trained model, or validated route planning.

The intended flow is to activate navigation, acquire environmental information, interpret context, and deliver an audio prompt. A working system would also need to announce when information is uncertain, unavailable, or stale. It should not present an unverified route as safe.

Prompts should prioritize immediate information without unnecessarily competing with environmental sound. Users should be able to pause assistance, repeat the last prompt, and stop the mode. Connectivity loss, camera obstruction, or processing failure should produce clear feedback rather than outdated guidance. Figure 5.2 preserves the original navigation mock-up.

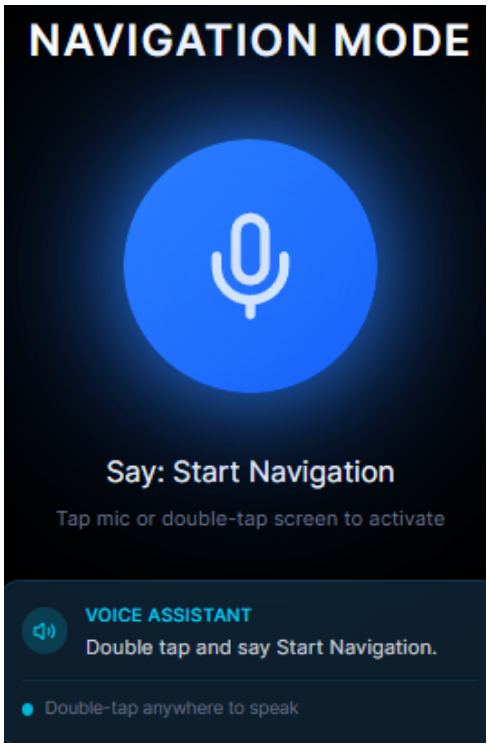


Figure 5.2: Original navigation interface. The microphone and double-tap instruction represent intended activation, not demonstrated navigation capability.

5.4 Voice-Based AI Queries

The query feature aims to make input review possible without reading a transcript. The intended sequence is to:

1. activate query mode through an accessible action;
2. speak a request and receive a listening-state cue;
3. hear the recognized query read back;
4. confirm, correct, or cancel it;
5. receive processing feedback and a spoken answer; and
6. request repetition, clarification, or detail.

Confirmation addresses the consequences of recognition errors, but does not itself improve recognition accuracy. Correcting input must also be distinguished from correcting a factually wrong answer: a correctly recognized query can still produce an unreliable response.

A future system could, for example, clarify whether "nine" refers to the number before proceeding. This illustrates intended dialogue rather than a transcript excerpt or test outcome. Figure 5.3 shows the query mock-up.

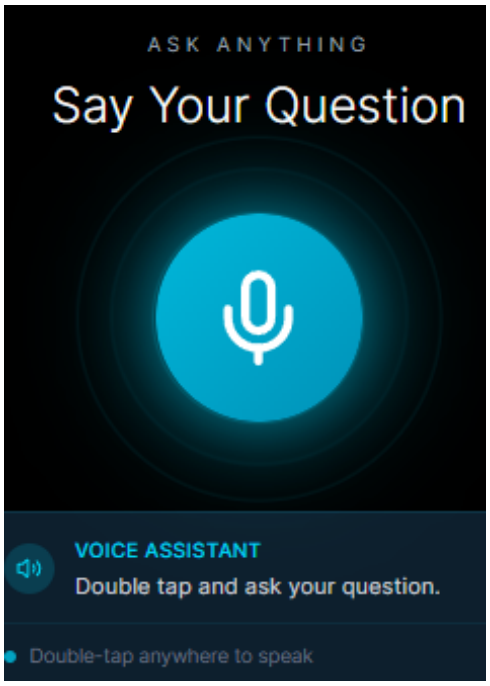


Figure 5.3: Original voice-query interface, representing a confirmation-oriented interaction concept.

5.5 Adaptive Audio Explanations

The explanation feature offers short and simple output for the key point and a detailed mode for additional context. It also proposes replay, saving, history retrieval, and voice-based style selection. Adaptation refers to user-selected presentation, not a validated personalization algorithm.

A summary should preserve important qualifications. Shortening an explanation must not remove uncertainty or make the system sound more confident than the evidence supports. Fluent spoken output also does not guarantee factual accuracy or faithful explanation of a model's internal reasoning.

A future implementation should distinguish response content, supporting information, and system limits. A contextual description could explain what was detected while stating that areas outside the camera view were not assessed. Figure 5.4 shows the original mock-up.

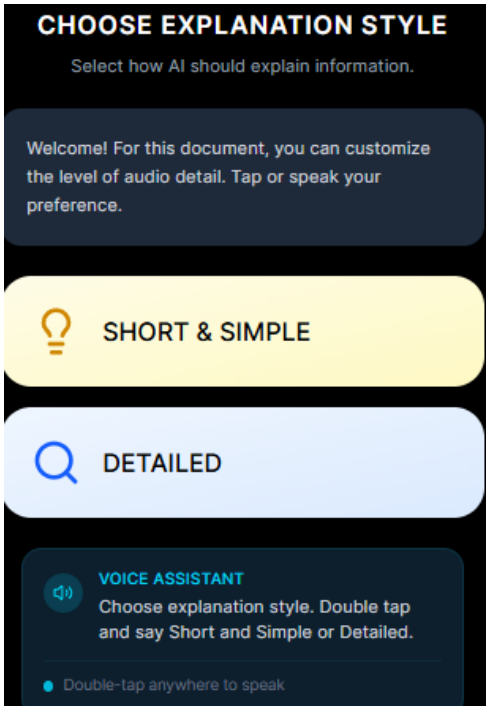


Figure 5.4: Original audio-explanation interface showing proposed presentation and playback controls.

5.6 Prototype Development

5.6.1 Paper Sketching

Early sketches explored navigation, voice queries, and explanation controls. Their purpose was to compare workflows before committing to visual detail. Figures 5.5 and 5.6 retain the paper concepts with their original labels and image quality.

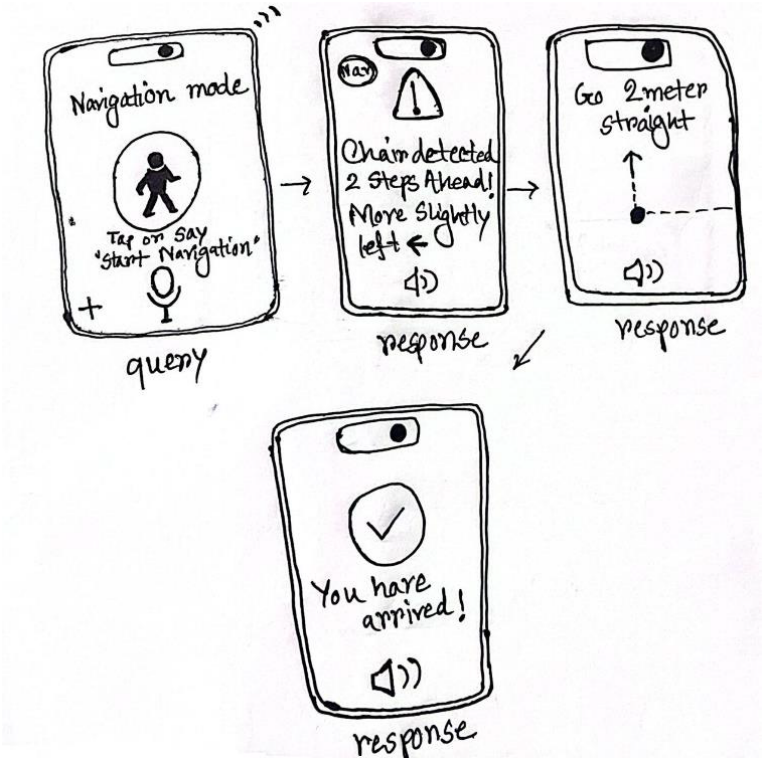


Figure 5.5: Original paper sketch for the navigation concept; an early design artifact rather than an implemented system.

5.6.2 Peer Review and Higher-Fidelity Design

Peer feedback emphasized reducing clutter, removing unnecessary visual instructions, and making confirmation explicit. These suggestions informed higher-fidelity screens and an interactive Figma prototype. Peer review supported iteration but did not replace testing with intended users or establish accessibility conformance.

5.6.3 Prototype Capabilities

The prototype represents screens, transitions, and intended action sequences. It supports discussion of discovery and interaction expectations. It does not establish a functioning speech service, AI backend, wearable device, audio pipeline, or accessible storage mechanism. Those components remain implementation requirements.

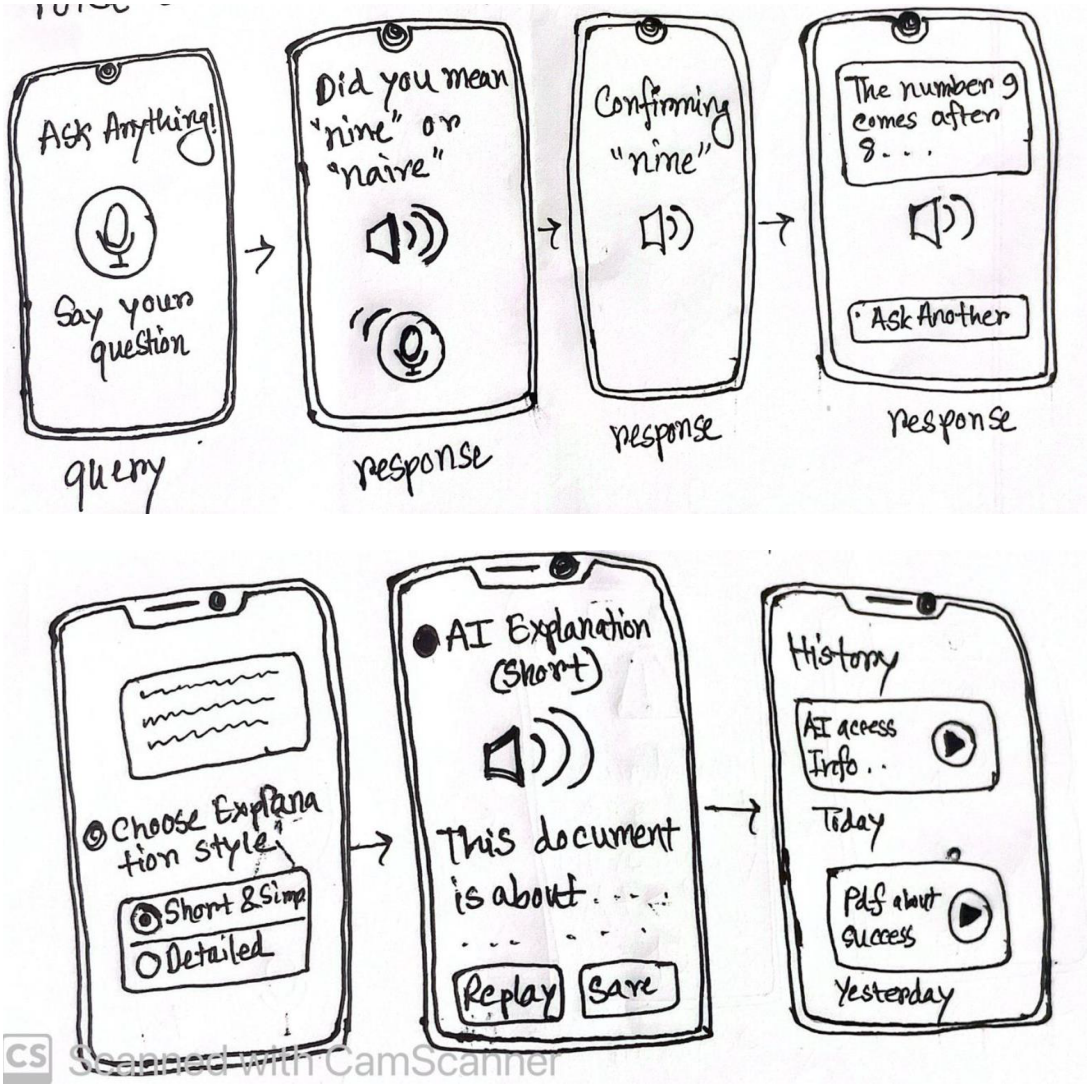


Figure 5.6: Original paper sketches for query and explanation concepts. No additional functionality is inferred from the drawings.

5.7 Gesture Model and Remaining Ambiguity

The documented redesign uses single tap to replay an explanation and double tap to activate a feature, confirm a query, or save an explanation depending on context. Voice commands select features and explanation styles. These mappings reduce dependence on locating small controls but remain mode-dependent.

The same gesture should not silently perform unrelated actions. Implementation should announce the active mode, expose available actions through accessible semantics, and confirm consequential actions. It must preserve operating-system screen-reader gestures; compatibility with VoiceOver or TalkBack requires direct testing.

A useful next refinement is to preserve a stable accessible activation action while separating commands such as “confirm query” and “save explanation.” This is a recommendation, not a separately evaluated change.

Chapter 6

User Evaluation and Redesign

6.1 Evaluation Procedure

The evaluation used an assisted Figma walkthrough and semi-structured discussion lasting approximately 30–45 minutes. One visually impaired participant is documented in the detailed methods. The team explained the prototype, observed interaction, and invited feedback.

Assistance helped the participant engage with the visual prototyping tool but limits conclusions about independent use. The record does not specify every intervention or how speech and audio were simulated. Results are therefore feedback on the interface concept and walkthrough experience.

6.2 Evaluation Tasks

Table 6.1: Task scope and interpretation of the evaluation.

Task	Intended focus	Supported interpretation
Navigation scenario	Activate guidance and understand prompts	Discovery and conceptual clarity; not mobility safety
Voice-query scenario	Activate input and review confirmation	Control access and expectations; not recognition accuracy
Explanation scenario	Select detail, replay, save, retrieve	Playback and history usability; not measured comprehension gains

No task completion rates, timing records, standardized usability scores, or quantitative workload results are reported. Such figures would require new data collection.

6.3 Usability Findings

6.3.1 Difficult Control Discovery

The participant had difficulty locating and distinguishing multiple controls. The source account associates this with hesitation, remembering positions, and confusion while switching features. The underlying problem is dependence on visually discovering controls rather than the mere presence of buttons. Properly labelled controls with predictable focus can remain useful in non-visual interaction.

6.3.2 Inconsistent Interaction Methods

Different features required different activation methods, making the interface difficult to predict. The participant had to work out how each feature operated. Both actions and system states need consistency, without hidden changes in a gesture's meaning.

6.3.3 Limited Feature Access

Some functions could not be activated easily when needed. The interface placed too much responsibility on remembering where and how to access them. This motivated broader activation surfaces, voice access, clearer onboarding, and feedback.

6.4 Changes After Evaluation

The source report describes four redesign responses:

- **Navigation:** simplify activation with full-screen touch interaction and an optional voice trigger.
- **Queries:** replace small-control dependence with gestures, add confirmation, and provide step-by-step audio feedback.
- **Explanations:** introduce tap-based replay and saving, voice-selected styles, and voice-accessible history.
- **Onboarding:** add a startup assistant explaining features, gestures, and voice commands.

These revisions respond directly to reported difficulties. The available record does not document a separate post-redesign study. Improvements in efficiency, confidence, workload, and independence remain intended outcomes rather than demonstrated effects.

6.5 Evaluation Summary

The evaluation's strongest contribution is problem identification. It exposed a gap between an interface described as voice-first and an interaction still dependent on visual controls. The redesign attempts to close that gap, but assistive technology compatibility remains unresolved. A functioning implementation and unassisted evaluation are needed before broader usability claims are justified.

Chapter 7

Ethical AI and Accessibility Requirements

7.1 Autonomy and Meaningful Control

Assistance should increase the user's ability to choose actions. The interface should offer an accessible stop action, input correction, and control over explanation detail. Users must understand when the system is listening and when it is inactive.

Autonomy also requires calibrated trust. Reassuring wording is inappropriate when the system cannot assess a situation reliably. A missed navigation hazard is particularly consequential: sensing limits should be communicated without presenting silence as evidence of safety.

7.2 Privacy and Consent

Voice queries, explanations, camera images, and location can reveal routines or third-party information. Implementation should collect only what an active task needs, identify remote processing, and obtain permission before saving history.

Microphone and camera activation should have understandable cues. Users should be able to review and delete saved material accessibly. Continuous recording should not be assumed necessary for voice activation. Camera assistance also raises bystander privacy concerns requiring explicit deployment and data-handling decisions.

7.3 Fairness and Inclusive Participation

The case does not represent the diversity of blindness and low vision. Needs may vary with language, accent, age, AT experience, additional disabilities, and context. Future work should include these differences and give participants a substantive role in setting priorities.

For a Bangladesh-based implementation, Bangla and English speech, code-switching, local pronunciation, and local navigation contexts are candidate test conditions. These are proposed directions, not demonstrated prototype capabilities.

7.4 Reliability, Safety, and Accountability

The navigation concept requires a documented boundary between information and safety-critical guidance. Before mobility testing, the team would need hazard analysis, controlled conditions, stopping rules, and appropriate safety oversight. Standard mobility aids and professional guidance should not be displaced on the basis of this prototype.

Users also need a way to report incorrect answers and inaccessible behaviour. Failure reports

should capture relevant system states without unnecessarily retaining sensitive data. Risk ownership and deployment decisions should be explicit.

7.5 Proposed Risk Register

Table 7.1 is an author-developed planning tool for future implementation, not a completed safety assessment.

Table 7.1: Proposed risks, safeguards, and validation activities.

Risk	Proposed safeguard	Validation
Missed or delayed obstacle information	Announce uncertainty; maintain stop control	Controlled hazard and latency testing
Incorrect speech interpretation	Read-back, correction, cancel	Varied speakers and noise conditions
Unexpected recording or retention	Capture cues, minimal storage, deletion	Permission and data-flow review
Screen-reader gesture conflict	Native accessible controls and alternatives	VoiceOver and TalkBack testing
Misleading explanations	Preserve qualifications; separate inference	Accuracy and understanding review

Chapter 8

Discussion, Limitations, and Future Work

8.1 Answers to the Research Questions

RQ1: Assistive workflows. Established AT use does not automatically make a new AI interface usable. Compatibility, control discovery, and accessible state feedback emerged as requirements. The study does not measure their prevalence across AI products.

RQ2: Speech and explanations. Misrecognized input and lengthy spoken responses were reported as sources of effort. Confirmation, repair, chunking, and replay follow from these concerns; their effects on performance and workload are unquantified.

RQ3: Feedback-informed design. The walkthrough identified visual-control dependence, inconsistent activation, and difficult feature access. These informed broader touch interaction, audio feedback, history, and onboarding. Screen-reader compatibility and gesture consistency remain to be tested.

RQ4: Contextual navigation. The participant's interest identifies a meaningful opportunity. A dependable solution requires sensing, timing, uncertainty handling, and supervised validation beyond the prototype. This study provides requirements rather than proof of navigation effectiveness.

8.2 Implications for HCI

The case illustrates why accessibility must be evaluated through complete tasks. A large microphone icon may communicate intent to a sighted reviewer while providing no usable path for a screen-reader user. Audio is insufficient if the user cannot interrupt, repeat, or correct it.

Consistency includes action meanings as well as physical gestures. Reusing double tap can appear consistent while introducing mode errors if its effect changes between activation, confirmation, and saving. Accessible feedback and native conventions are essential companions to simplified controls.

Explanation quality should also be judged against the user's goal. The interface should support both brevity and inspection, allowing users to obtain a key point and request detail without losing important qualifications.

8.3 Study Limitations

- **Small sample:** one participant is documented in each stage, with uncertain overlap. Generalization across blind and low-vision populations is not supported.
- **Incomplete records:** transcripts, a codebook, intervention logs, and a full analytic audit trail are unavailable in the report.

- **Assisted evaluation:** facilitator support may affect performance and feedback; independent usability was not established.
- **Prototype fidelity:** Figma represents flow, not reliable speech, perception, or production accessibility.
- **No comparative measures:** timing, error rates, comprehension, workload, and baseline results were not reported.
- **No documented retest:** the redesign was not separately evaluated in the available record.
- **Ethics documentation:** consent, recording permissions, retention, and review status are not established by the supplied material.

8.4 Future Work

The next step should be a functioning prototype with native accessible controls and clear audio state management. Query confirmation and explanation playback can be tested before introducing navigation, separating general interaction questions from mobility risks.

A subsequent study should recruit participants with varied visual conditions, AT experience, and language preferences. It should distinguish independent from assisted completion and document each intervention. Appropriate outcomes include task success, correction attempts, completion time, comprehension, perceived workload, and qualitative preferences. Target values should be set prospectively.

Navigation research should proceed through controlled sensing tests and supervised indoor scenarios with explicit failure handling before less predictable environments are considered. Deployment requires evidence about missed hazards, delays, uncertain descriptions, connectivity loss, and whether audio interferes with environmental awareness.

Chapter 9

Conclusion

This project investigated accessibility concerns through a formative interview and used them to develop an AI-assistance interface concept. Reported needs concerned assistive workflow compatibility, speech repair, manageable spoken explanations, and contextual information in unfamiliar environments. These informed navigation, query, and explanation features in a Figma prototype.

An assisted single-participant evaluation exposed problems with control discovery, consistency, and feature access. The redesign responded with broader activation surfaces, contextual gestures, spoken confirmation, explanation management, and onboarding. These provide a direction for implementation, but the evidence does not establish improved workload, recognition accuracy, independent mobility, or safety.

The central contribution is a traceable design argument: accessible AI requires usable controls, understandable feedback, recoverable errors, and meaningful choice. Future work should preserve that focus while testing a functioning system, documenting ethical procedures, and including a wider range of users.

References

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Appendix A

Recommended Follow-up Study Materials

These materials are proposed for future use. They are not the original script, evidence of completed procedures, or additional results.

A.1 Accessible Session Checklist

1. Provide study information in the preferred accessible format and confirm voluntary consent.
2. Ask about AT, input, audio settings, and accommodations.
3. Explain which features are operational and which are simulated.
4. Ask the participant to locate a feature, submit and correct a query, replay an explanation, and retrieve a saved item.
5. Record independent, assisted, incomplete, or abandoned tasks and every intervention.
6. Ask about clarity, control, effort, trust, and privacy without implying a preferred response.
7. Explain withdrawal and deletion options and provide an accessible debrief.

A.2 Suggested Interview Prompts

- How did you know the system was ready to listen?
- What did you expect the selected action to do?
- How would you correct a misunderstood request?
- Which information was useful, and which could be omitted?
- When would you want a short answer or more detail?
- Could you stop, repeat, or leave when you wanted?
- What would you need to know before allowing audio, camera, or location processing?

A.3 Suggested Evidence Record

For each task, record the goal, device and AT settings, completion status, assistance, correction attempts, relevant states, and comments. Store identifying information separately from research notes. For functioning AI studies, record software and model versions so failures and changes can be investigated.